KLE Technological University | Creating Value, Leveraging Knowledge

Challenge League

BUILDVERSE

IDEATE · ENGINEER · OPTIMIZE

Buildverse

Ideate • Engineer • Optimize

BUILDVERSE is an engineering innovation challenge where teams transform an idea into a functional prototype using limited resources. Teams must strategically procure electronic and mechanical components using a predefined points system, design an innovative solution, and build a working prototype.

Participants must balance engineering creativity, strategic planning, and resource management while optimizing their prototype through multiple improvement cycles. The goal is to build the most innovative and functional prototype while minimizing point expenditure.

Objectives of the Event:

The event is designed to:

- Encourage innovation and creative problem solving
- Promote hands-on engineering and prototyping skills
- Develop strategic resource management
- Improve rapid prototyping ability
- Foster team collaboration and engineering thinking
- Participants will simulate real-world product development where engineers must build solutions with limited resources and time constraints.

Eligibility

- Students from any branch are eligible to participate
- Basic knowledge of electronics / Arduino programming is recommended

Team Composition

- Maximum Teams: 20
- Team Size: 4 members per team

Team Rules:

- Teams must register before the event deadline.
- No team member changes will be allowed after the event begins
- Each participant may belong to only one team

Event Structure / Rounds

Round 1: Component Analysis & Strategy Planning
Teams will receive:

- Component catalog
- Component pricing (points)

Event rules

- Teams must analyze the available components and develop a design strategy.
- Duration: 10 minutes

Round 2: Procurement Round

- Teams receive a **fixed number of points (1000 points).**
- Teams must use these points to **purchase required components** from the component catalog.
- Teams must carefully plan purchases because points are limited.
- Duration: 15 minutes

Round 3: Prototype Development – Phase 1

Teams will assemble and program their prototype using purchased components. Activities include:

- Circuit building
- Mechanical assembly
- Programming
- Testing
- Duration: 90 minutes

Round 4: Optimization Cycle

Teams may improve their prototype through a market-style optimization phase. Teams can:

- Exchange components
- Purchase additional components
- Sell unused components
- Use points to obtain mentor assistance

This cycle may repeat 2–3 times to improve the prototype. Duration: 60 minutes

Round 5: Final Build Phase

- Teams finalize their prototype by implementing improvements and completing functionality.
- Duration: 60 minutes

Round 6: Final Presentation and Evaluation

Each team will present their prototype to the judging panel.
Teams must explain:

- Concept
- Design approach
- Functionality
- Resource strategy
- Duration: 30 minutes

Event Flow

Registration
↓
Event Briefing
↓
Component Chart Distribution
↓
Strategy Planning
↓
Procurement Round
↓
Prototype Development Phase 1
↓
Optimization Cycles
↓
Final Build Phase
↓
Prototype Presentation
↓
Judging and Results

Mentor Assistance System

Teams may use their points to obtain technical guidance.

Assistance Type	Cost
Basic Hint	20 points
Circuit Guidance	40 points
Code Debugging	40 points

This system encourages teams to balance independence and assistance

Participant Responsibilities

Participants must:

- Handle components carefully
- Return borrowed components if required
- Follow safety guidelines
- Maintain discipline during the event
- Respect mentors and judges

Rules and Regulations

- Teams must follow all instructions given by the organizers. Only components purchased through the point system may be used.
- External components are strictly prohibited. Teams must manage their points strategically.
- Any damage to equipment due to negligence may lead to penalties.
- Teams must complete their prototype within the allocated time.
- All components provided during the event must be returned to the organizers after the event concludes.
- Exchange or sharing of components between teams is strictly prohibited.

Judging Criteria

Total Score: 100 Marks

Criteria	Marks
Innovation	30
Functionality	30
Engineering Design	20
Resource Efficiency	20

Time Constraints

Stage	Marks
Registration & Briefing	30 min
Component Analysis	10 min
Procurement Round	15 min
Prototype Build Phase 1	90 min
Optimization Cycle	60 min
Final Build Phase	60 min
Presentation & Judging	30 min

Safety Guidelines

- Avoid short circuits
- Handle electrical components carefully
- Ensure proper wiring
- Follow mentor and organizer instructions

Tie-Breaking Rules

In case of a tie:

- Higher Innovation Score
- Higher Resource Efficiency Score
- Final decision by Judging Panel

Organizer Authority Clause

- The decisions of the judges and organizing committee will be final and binding.
- Organizers reserve the right to modify rules if necessary for smooth execution of the event.

Additional Event Specifications

Initial Points Allocation

- Each participating team will be allocated 1000 points at the beginning of the event.
- Teams must use these points strategically to procure components, request mentor assistance, and optimize their prototype.

Component Return Policy

- Teams must return unused components to the component store present in the venues during optimization rounds.
- Returned components will be credited at 50% of their original point value.
- Components must be unused and in proper condition to be eligible for return.
- This mechanism encourages strategic decision-making and resource optimization.

Laptop Requirement

- Teams must bring their own laptops for programming, debugging, and development purposes.
- Organizers will not provide laptops during the event.
- Participants are also responsible for bringing any necessary software, drivers, and development environments required for their prototype.

Expected Outcomes

Participants will gain:

- Hands-on prototyping experience
- Practical engineering design thinking
- Strategic decision making
- Strong team collaboration skills

Contact Information

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